Agreeing With the Frame: sycophancy and psychosis in an agent that judges companies

Yuting FU Reflection October 2026

Two failure modes are worth naming separately. **Sycophancy** is the model agreeing with the frame it is given — the user's premise, the confident document, the flattering reading. **Psychosis** is subtler: the model's own generated narrative becomes its evidence, and a story that is internally consistent starts to function as if it had been checked. Building a Fisher agent forced me to notice that both are failures of *provenance*, not of intelligence, and that neither can be fixed by asking the model to be careful.

01 Sycophancy has a shape, and this homework is built around it

The assignment's data layout is unusually honest about the problem. The three evidence sources are not equal. The annual letter, the four earnings calls and the news file are *management's own words*, and the homework says plainly that they "are not evidence on their own: check what management claims against the database and the interviews". A shareholder letter is, structurally, a sycophantic document: it is written to be agreed with. A verifier that reads it and nods has done nothing.

The interviews make the same point from the other side: only some roles know each topic, everyone else says they cannot comment, and "people who know always agree, so one good answer is enough". That is a strange property for a research task — normally you would sample opinions and weigh them. Here the discipline is to work out *who is entitled to speak* and then stop. The model is not asked what it thinks; it is asked to find the right source.

Once I framed it that way, the dangerous question became obvious. If I had written a prompt like "is a 9.5% operating margin worthwhile for this industrial-robotics company?", the model would have answered inside my frame: yes, on a competitive industry, with that capex cycle, for a business of that size. Every one of those qualifiers would have been invented to support the number I had already put in the sentence. The verdict would have looked like analysis and would have been a mirror.

> *I removed the opportunity rather than the tendency. My agent never asks the model a question it can be sycophantic about: the three verdicts are computed in code, and the model receives finished numbers with an instruction not to recompute them.*

02 Psychosis: when the story becomes the evidence

The failure I found harder to see is the one where nothing goes visibly wrong. Give a language model a set of numbers and ask it to explain them, and it will produce a coherent account — margins fell because input costs rose and the new plant is still ramping; growth slowed because the market matured. These explanations are not lies. They are also not derived from anything. They are the shape a plausible paragraph takes, and they arrive with the confidence of a finding.

The homework's own warning is the measurable version of this: "arithmetic done by the model was the most common cause of wrong verdicts". A model that computes a number slightly wrongly does not usually announce uncertainty — it produces the number and then a paragraph explaining it. The explanation is what makes the error expensive, because it is what a reader trusts. The loop closes when the model's output re-enters its own context as input: an agent that writes a summary, then reads that summary back as evidence about the company, has begun to hallucinate with perfect internal consistency.

FAILURE MODE	WHAT GOES WRONG	WHERE IT WOULD HAVE APPEARED HERE
Sycophancy	The answer inherits the frame of the question — the document, the prompt, the confident claim.	Reading the shareholder letter and accepting its account of margins; asking the model to judge "worthwhile" in the management's own vocabulary.

Psychosis	The model's own narrative becomes the evidence; coherence substitutes for verification.	Explaining a margin move instead of measuring it; letting a model-computed number stand because the surrounding sentence sounds right.

Both are cured the same way, and it is not prompt engineering. The answer has to come from somewhere the model does not control. In my agent that "somewhere" is the read-only database plus the published answer key: the growth rate, the two operating margins and the industry medians are computed by SQL, and the rules that turn them into PASS or FAIL are five lines of Python that the model never sees the inside of.

03 What the design had to give up, and what it gained

Choosing three points that live entirely in the database is a retreat, and I want to be honest about it. Points 4, 7, 9 and 11 — sales organisation, labour relations, depth of management, competitive edge — are the ones where Fisher's method is most alive, because they require scuttlebutt, judgement and the willingness to weigh a source. I avoided them because a model reading an interview answer is exactly where sycophancy enters: the answer is authoritative in tone, one sentence long, and the model has every incentive to accept it. A stronger submission would have used the interview tool and shown how to keep the model honest while doing so. I chose rules that a reviewer can re-derive with a calculator, and I would rather say that plainly than dress it up as a philosophy.

What the retreat bought is falsifiability. Every verdict can be recomputed by hand from the database, so when the public answer key disagreed with me, I had something to check — and I did disagree at first. My initial two-year growth was a compound annual rate; the key uses the simple two-year change. Once the definition was right, the rules reproduced all sixty verdicts on the twenty public companies, and the closest company still sits 3.1 percentage points from the industry median on growth and 1.7 points from it on margin. On the margin-trend point, every company's margin moves monotonically across FY2023–FY2025, so five different windows all give the same verdict. That is what a rule measuring the data looks like, as opposed to a rule measuring the answers.

04 Why this matters in finance

Fisher's fifteen points are a control framework before they are an investment method: each one asks for evidence of a specific kind, from a specific source, about a specific claim. That is also what KYC and equity due diligence are — the work is not to have an opinion about a company, it is to reconcile what a company says with what can be checked. An agent that only agrees with the documents has automated the least useful half of that job, and an agent that narrates its own reconciliation has automated the most dangerous half, because the output looks finished.

The conclusion I take from this homework is narrow but, I think, durable: the value of an agent in a verification task is not its judgement, it is the discipline of where its judgement is allowed to act. My agent uses a strong model and gives it almost nothing to decide — and that is the design decision I would defend, not the tools it calls. Sycophancy and psychosis are both what happens when a model is asked to be the place where the answer comes from. It should be the last step, never the source.

Written for FTEC5660/SEEM5660 Homework 3, Task 2. The figures quoted (60/60 verdicts, 3.1pp and 1.7pp nearest margins, monotone FY2023–FY2025 margin paths) are from my own runs on `public_test`; the rule definitions are in `hw3.py` and the report in `README.md`.